Regulation and Safety

Station BlackOut - Safety Analysis

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Introduction

The aim of this report is to present the first part of the Station Black-Out (SBO) project developed with RELAP5 as a system code. The main purpose of this first delivery is to study the reference model before any new safety-system design is introduced. In particular, the work is divided in two tasks: first, the verification of the steady-state conditions of the plant model and second the analysis of the SBO base case transient. The steady-state analysis is used to check that the model starts from physically consistent and stable operating conditions. The main thermal-hydraulic parameters are compared with the target values provided in the project statement, and their time evolution is checked in order to confirm that the initial condition is sufficiently stable for the transient calculations. After the steady-state verification, the SBO base case is analyzed. This case represents the reference accident scenario and is used as the starting point for the following parts of the project. The objective is to understand the sequence of events, the system response and the main thermal-hydraulic trends during the transient.

Methodology

The reference plant selected for this analysis is the Zion Nuclear Power Station (United States), which was decommissioned in 1998. Although the real plant originally had four primary loops, the RELAP model used in this project is simplified into two equivalent loops only. One loop represents the broken loop and keeps the dimensions of a single loop, while the other represents the intact side and groups the remaining three loops into one single equivalent loop.
Figure 1 shows the nodalization used in order to represent in the best way possible the Zion nuclear power plant.

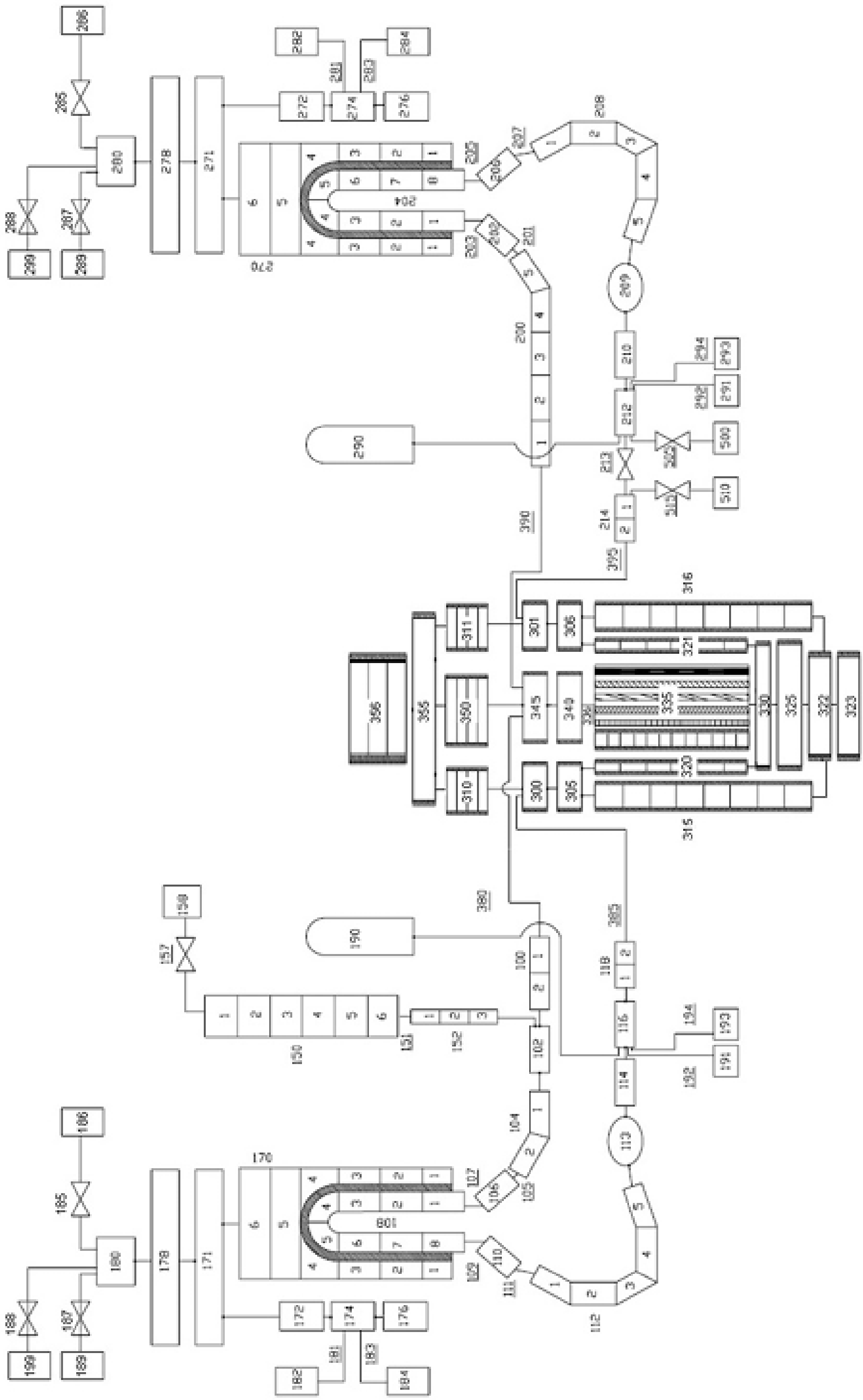

Figure 1: RELAP5 nodalization of the Zion Nuclear Power Plant model.

1 Steady State Analysis

Before starting the accident analysis, it is necessary to verify that the model is in a stable and realistic initial condition. A correct steady state is important because the transient results strongly depend on the quality of the initial condition. The steady-state run is checked by comparing the calculated values with the reference values and by observing the time evolution of the main parameters. If the variables remain approximately constant and the deviations from the target values are acceptable, the initial condition can be considered suitable for the transient simulation.

Parameter	Expected value	Steady State Results
Power (MW)	3250	3250.0
Primary pressure in the hot leg (MPa)	15.5	15.5
Secondary pressure (MPa)	6.7	6.71
Pressurizer level (m)	8.8	8.86
Broken Steam Generators DC level (m)	12.2	12.2
Intact Steam Generators DC level (m)	12.2	12.2
Core outlet temperature (K)	603	603.4
Core inlet temperature (K)	571	570.8
Broken Main feedwater flow (kg/s)	459	455.8
Intact Main feedwater flow (kg/s)	1325	1324.4
Intact loop flow (kg/s)	12865	12436
Broken loop flow (kg/s)	4351	4356

Table I: Expected vs Simulation results for Steady State Values

Table I shows the main variables that must be checked in the steady-state calculation. The steady-state calculation shows that the main plant variables are close to the reference values; small deviations may appear because of nodalization choices, control-system settings, and numerical approximations. However, the overall agreement is sufficient to accept the initial condition for the SBO transient analysis.

1.1 Steady State Verification

In this section some plots regarding the steady state case are provided. The purpose is only to ensure that the calculations done with RELAP are coherent with the steady state conditions the plant should be operating at.

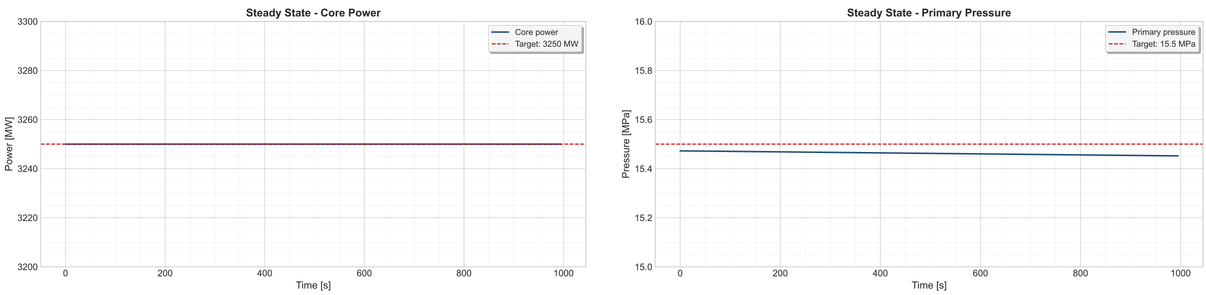

Figure 2: Core Power (on the left) and primary pressure (on the right).

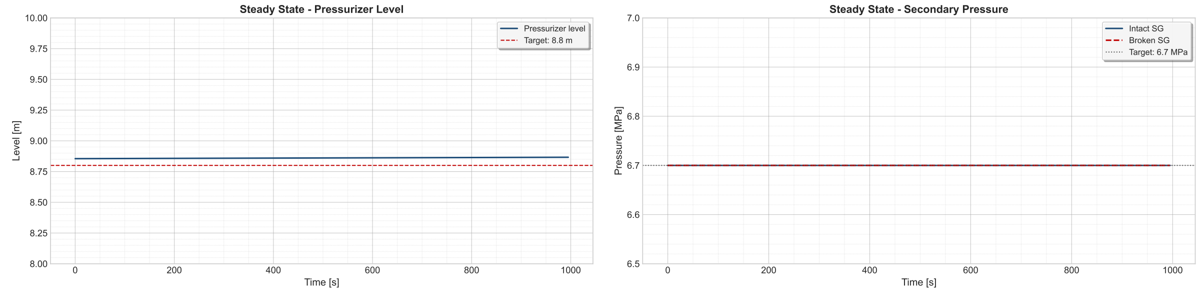

Figure 3: Pressurizer Level (on the left) and Secondary Pressure (on the right).

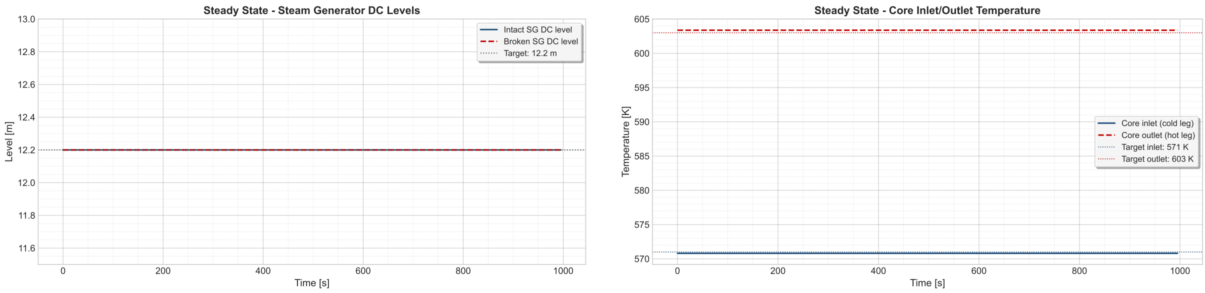

Figure 4: SG levels (on the left) and RCS temperatures (on the right).

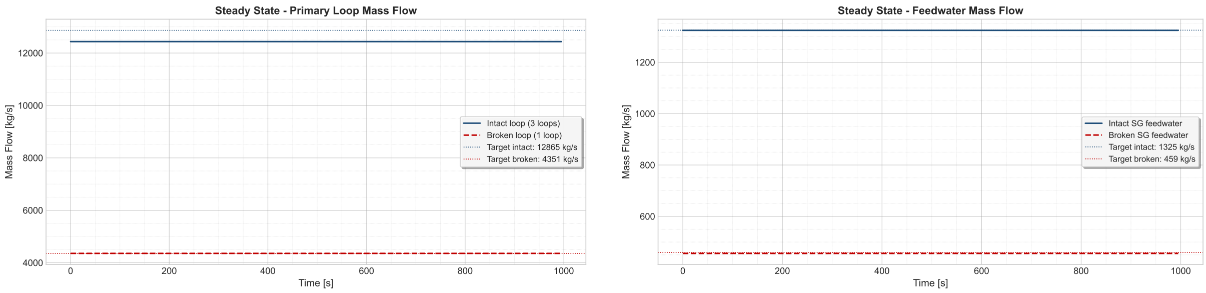

Figure 5: Primary Mass Flow (on the left) and FW Mass Flow (on the right).

As we can see, all the variables analyzed are stable and do not change over the time. Some of them present little uncertainties that are probably due to numerical errors and the complexity of the phenomena involved.

2 Base Case Analysis

After the steady-state verification, the SBO base case is studied. The Station Black-Out is assumed to begin at $t = 0$ s with aggravated conditions. Under these conditions, every system that depends on electrical power is considered unavailable. As a result, the reactor coolant pumps trip, the control rods are inserted, and other relevant functions, such as the cooling of the pump seals, are lost. The general assumptions that are common to all the considered accident sequences are summarized in Table II below.

Table II: Main events to be included in the SBO base case description.

Event / action	Short description
SBO initiation	Loss of off-site and on-site AC power according to project assumptions
Reactor scram	Automatic shutdown following the transient logic
RCP trip	Reactor coolant pumps stop after loss of power
FW / AFW behavior	Feedwater and auxiliary feedwater response according to the base-case logic
Pressurizer systems	Heater and control-system behavior during SBO
Safety injection unavailability	HPSI and LPSI unavailable during the transient
Steam line / relief actions	Valve actions according to the project assumptions
Additional operator actions	To be completed based on the adopted base-case procedure

In order to simulate all the accidental sequence, some actions that could be performed from the operators have been taken into account. These are listed below.

Action / Event	Description
RCP Seal Leak	Starts 10 minutes after SBO initiation with a leak size of 0.00001 m2 per pump.
Leak area increase	The leak area increases 4x when saturation takes place at the RCP.
TDP AFW Injection	Auxiliary feed water is provided by a turbo-driven pump (TDP) which is controlled manually and requires batteries.
TDP battery depletion	Batteries last for 5h, after which the TDP becomes uncontrolled, injecting water until the separator is flooded and the TDP stops.
Cooldown strategy (EOP)	Operators follow a strategy to cool down the SG at a rate of 55 K/h until 15 bar.
Primary depressurization (EOP)	Operators fully open the Pressurizer relief valves to depressurize the primary system if the Core Exit Temperature is higher than 923 K.
AC power recovery	Recovery of AC power occurs after 12h, which recovers AFW, HPIS, and LPIS.

Table III: Configurable actions and conditions for the SBO sequences in the Zion model

2.1 First Analysis

The first analysis was performed considering that all the EOPs listed in Table III have taken place. We are going to analyze the transient in order to verify the situation of the plant after the SBO has occurred.

2.1.1 Reactor Coolant System

Right after the SBO event, the SCRAM immediately shuts down the reactor by inserting the control rods, leading to a rapid decrease in power, that is plotted in Figure 6. The residual value present in the graph is due to the decay heat, which represents the main issue in this types of scenarios. Moreover, the loss of AC power causes the immediate trip of the pumps.

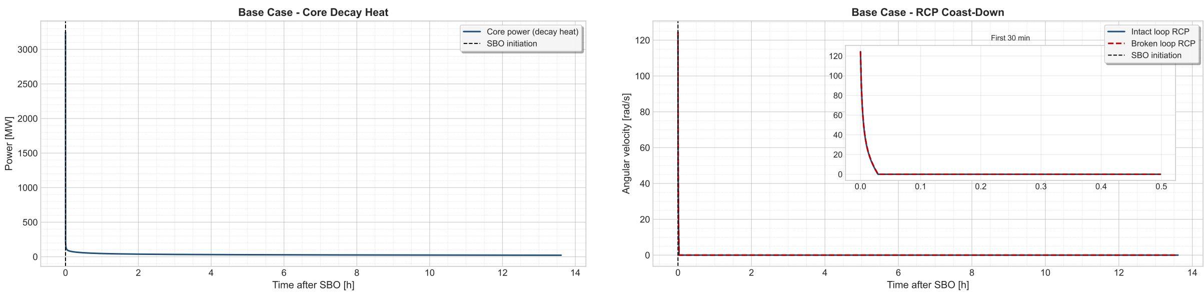

Figure 6: Core Power (on the left) and RCP velocities (on the right).

The sudden power decrease causes a contraction within the coolant, which lowers the level inside the pressurizer, as it is shown in Figure 7. The level reaches zero and it starts to increase again after 12 hours, when the AC power is recovered and water is introduced again by the means of HPIS, LPIS and the pumps. The loss of level is also enhanced by the loss of inventory, as the seal injection system is not available during a SBO. This lead to some leaks from the pumps, which contributes to the decrease. However, an empty pressurizer does not imply that the core is uncovered. This situation is properly described in Figure 7.

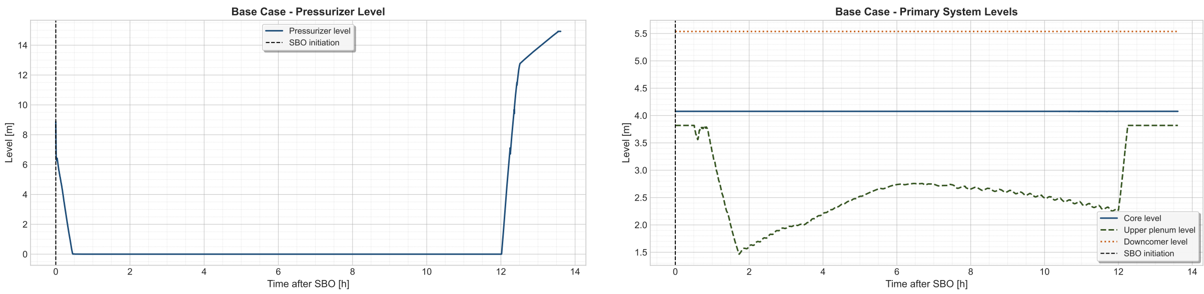

Figure 7: Pressurizer Level (on the left) and Core Levels (on the right).

In Figure 8 the pumps's leaks is shown.

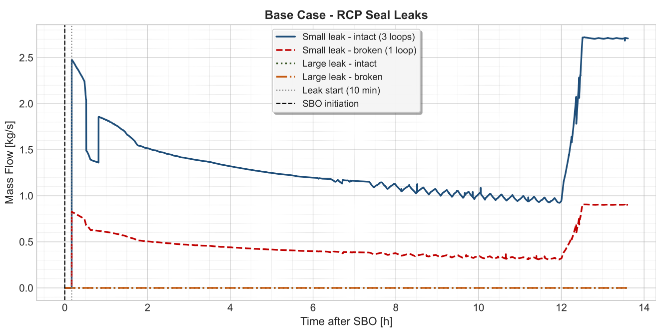

Figure 8: RCP Seals Leaks. Notice that the intact loop represents 3 loops.

According to the EOPs, if the outlet temperature from the core is higher than 923 K, the operators should depressurize the primary system using the PRZ valves. In our case, this procedure has not been necessary, as the core exit temperature does not overcome 923 K. Figure 9 shows the main parameters related to the temperatures of the core.

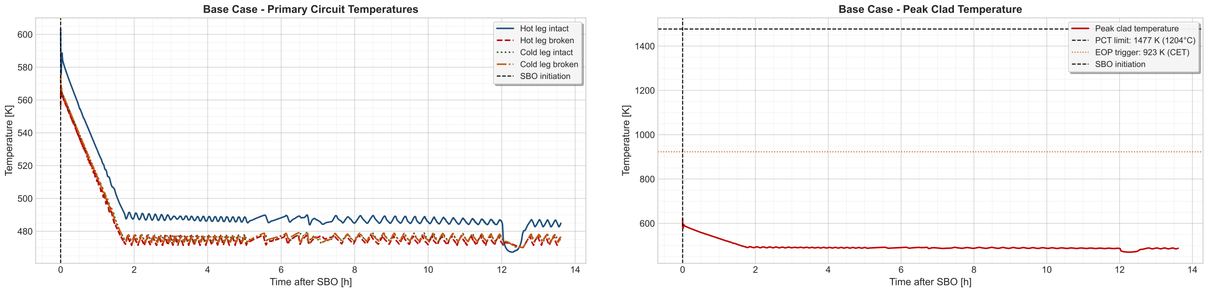

Figure 9: Core Temperatures (on the left) and Cladding Temperature (on the right).

The primary temperatures are lower than the threshold limit established for the operator, as well as the peak cladding temperature. It remains lower than the safety limit, preventing damages to the cladding. It also indicates that DNBR is prevented and the core is covered. The constant behaviour can be explained as a consequence of some natural circulation phenomena that could have taken place inside the primary, due to the pressure difference between primary and secondary. Figure 10 shows that a little flow inside the RCS is present, even if the pumps are tripped. The natural circulation guarantees the heat exchange between the core and the SGs.

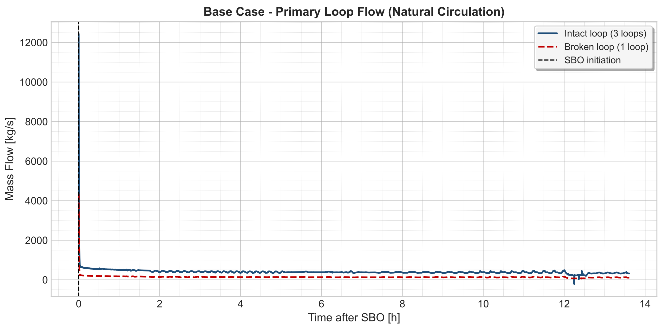

Figure 10: Primary Mass Flow

2.2 Secondary System

It is finally the moment to take a look at the behaviour of the secondary system. In a SBO scenario, this system plays a fundamental role, since it is used to extract the heat coming from the core where no AC powered system is available. A natural circulation flow regime is established by the pressure's difference between the primary and the secondary, helped by the elevation of the latter. In fact, the SG is usually located in a higher position than the core, in order to stimulate the gravity forces that act by density.
As we said in the previous chapter, a cooldown procedure is performed by the operators. The aim is to cool down the SG at a rate of 55 K/h, until a pressure of 15 bar (1.5 MPa) within the secondary is guaranteed. In Figure 11 this situation is well described.

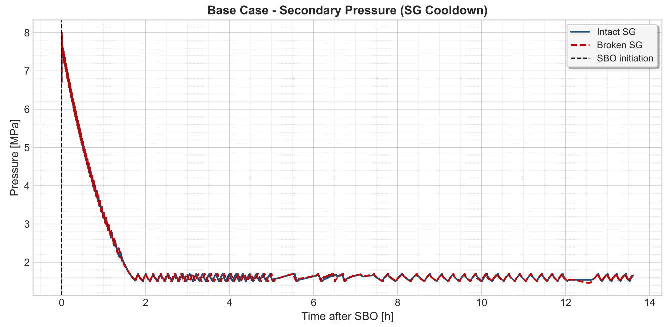

Figure 11: SG Pressure Evolution during the Transient.

The procedure is done by the relief valves, located at the top of the SG. Their function is to release steam to the atmosphere, in order to reduce the pressure inside the SG. As we can see from the picture, the procedure is well carried out until the target pressure of 15 bar is reached.

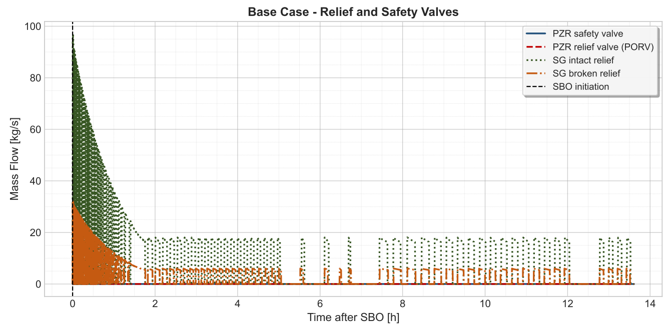

Figure 12: Relief Valves Flow. Notice that the flow coming from the PRZ valves is zero, as the primary depressurization has not been carried.

Figure 12 shows the flows coming from the relief valves. The behaviour of the valves is coherent with the SG pressure curve, the most of the steam is released at the beginning of the procedure and gradually decreases. After that, the flow remains constant, since the operators aimed to keep the SG pressure constant and the Auxiliary FeedWater is constantly injecting water. The latter is shown in Figure 13. The batteries that power the turbo-driven AFW last for 5 hours. As mentioned in Table III, at this moment the TDP becomes uncontrolled and injects water until the SG separator is flooded.

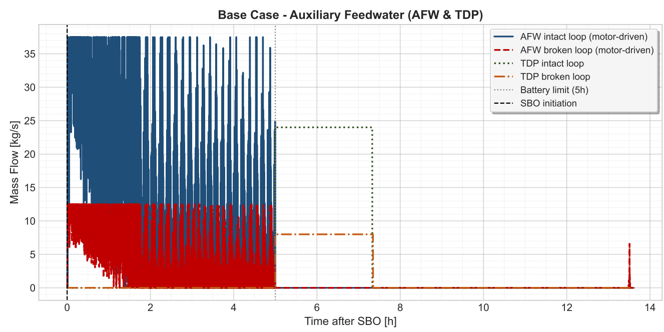

Figure 13: Auxiliary Turbo-Driven Feedwater Flow. Remember that the intact loop represents 3 loops.

3 Additional Calculations

As part of the base case simulation, there are two operator procedures assumed to be preformed. These are the steady depressurization of the secondary system, and the depressurization of the primary based on a high outlet core temperature. Apart from the complete base case, the simulation was also performed without these operator interventions in order to gauge their effectiveness. The second EOP mentioned above, pertaining to primary depressurization, was never actuated in the base case as the outlet temperature never reached the threshold value. Therefore, this case was identical to the base case and will not be analyzed further and only the additional case without the EOP 1 is analyzed below. This simulation analyzes the dynamics and critical values of the system when the steam generators are not depressurized allowing for a further reduction in primary temperature and pressure.
Figure 14 below demonstrates the lack of pressure relief of the secondary system. The lack of the depressurization EOP maintains the pressure high in the secondary system. This graph verifies that the EOP is not initiated as intended. In the base case, the relief valve setpoint is progressively reduced from the nominal value of 6.7 MPa down to 1.5 MPa, following the 55 K/h cooldown ramp. Without this operator action, the setpoint remains fixed at the nominal opening value of 7.58 MPa throughout the entire transient. As a consequence, the secondary pressure does not decrease monotonically but instead exhibits a strong cyclic behaviour: as the decay heat continuously produces steam, the pressure rises until the relief valve opens at 7.58 MPa, releasing steam to the atmosphere. Once the pressure drops below the closing setpoint of 7.79 MPa, the valve closes and the pressure begins to rise again. This open-close cycling of the relief valves repeats continuously, maintaining the secondary pressure oscillating between approximately 4.5 and 8.0 MPa rather than reaching the 1.5 MPa target of the EOP.

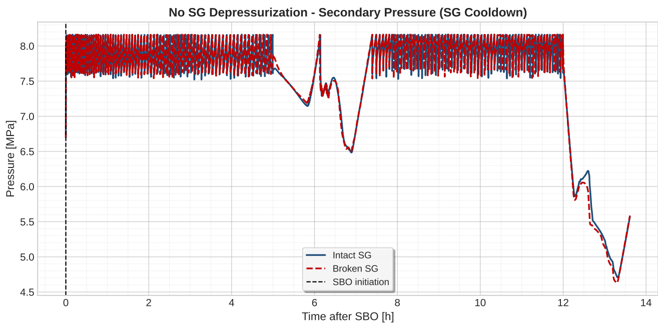

Figure 14: Secondary Pressure.

One of the main concerns during the transient event is the protection of the fuel cladding. Figure 15 shows the temperature of the cladding as a way to verify a lack of oxidation of the zirconium. The temperature in the primary does not reach excessive levels. Although it is higher than the base case which sees a drop in temperature of the cladding shortly after the event initiation, the temperature is still a safe margin away from rapid oxidation.

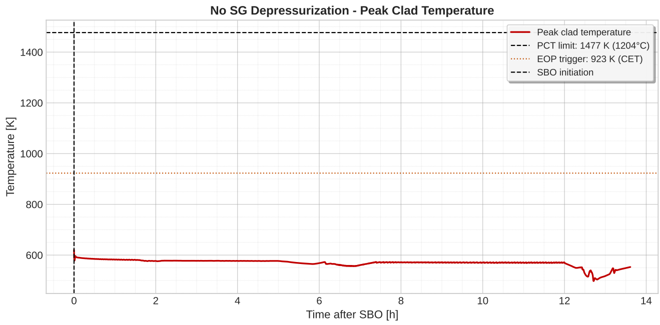

Figure 15: Peak Cladding Temperature.

Another factor of concern is the pressure in the primary. Without the further cooldown of primary system as a result of the steam generator depressurization, the pressure in the primary is much more elevated. One of the main concerns of this value is the correlation between leak rate and pressure. With higher primary pressures, we can expect to lose more inventory through the RCP seals. This situation is well represented in Figure 16.

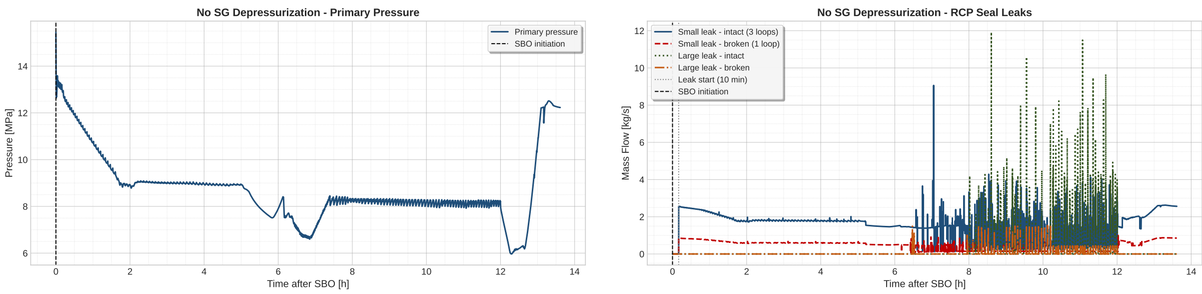

Figure 16: Primary pressure (on the left) and RCP seal leak rate (on the right).

As discussed above, the leak rate of the RCP seals is of concern in this event, and becomes more of an issue with higher primary pressures. Figure 16 confirms this. The higher primary pressure has a direct consequence on the RCP seal leak rates. Since the leak flow is driven by the pressure difference between the primary circuit and the containment atmosphere, a higher primary pressure results in a significantly larger leak flow. The seal leak rates in the no-cooldown case reach peak values of approximately 8-10 kg/s for the intact loop, compared to roughly 2.5 kg/s in the base case — about 3-4 higher. The irregular oscillating pattern of the leak rate directly correlates with the cyclic pressure oscillations of the secondary system described above. Despite this increased inventory loss, no core uncovery is observed within the 12-hour simulation window, indicating that the plant remains in a safe condition. However, in a longer SBO scenario, the sustained high leak rate would represent a more serious concern for primary inventory management.
The simulation demonstrated that the EOP does help to reduce the temperature of the core and the pressure of the primary system. However, based on this analysis is not essential in order to protect the integrity of the fuel cladding during a 12 hour station black out.

4 Event tree for the SBO scenario

The event Tree analysis is an inductive procedure that aims to show all the possible outcomes of an initiating event, in our case an SBO with aggravated conditions. The objective is to identify all the potential accident scenario in a complex frequency. At the end of the process, it is possible to calculate the core damage frequency by assigning a frequency to each sequence. In this part an event tree for the SBO scenario regarding the ZION nuclear power plant is presented, together with the PSA frequency calculation.

4.1 Initiating Event and Top Events considered in this analysis

The initiating event is the SBO occured in aggravated conditions. It implies that all the systems that require power are unavailable. The top events that were chosen to be analyzed in this event tree are listed in Table IV.

Table IV: Top events considered in the SBO event tree.

ID	Top event	Description	Timing	Success criterion
TE1	Small leak	RCP seal LOCA induced by loss of seal injection and thermal barrier cooling. Break area 1×10^{-5} m^2 per pump (intact loop counts as three pumps).	$t = 10$ min	Seal integrity maintained; no leak develops.
TE2	AFW & TDP	Turbine-driven pump delivers auxiliary feedwater to the steam generators, providing the only available secondary heat sink during SBO.	$t = 0^+$ (on demand)	TDP starts and supplies sufficient feedwater to both SGs.
TE3	Large leak	Amplification of the seal leak by a factor of four when saturation conditions are reached at the RCP location, according to NUREG/CR 7110 Vol. 2.	$t = t_{\mathrm{sat,RCP}}$	Leak area does not amplify (or saturation not reached at the seals).
TE4	Extra batteries	DC power supply enabling manual control of the TDP and key instrumentation. Beyond depletion the TDP runs uncontrolled until the SG separator floods and trips it.	$t = 5$ h	Batteries provide controlled TDP operation until AC restoration.
TE5	AC restoration	Recovery of off-site or emergency on-site AC power, restoring AFW, HPSI, LPSI and component cooling.	$t = 12$ h	AC power restored before core uncovery.

Chosing 5 top events for the event tree would yield 32 possible sequences. However, some sequences can be discarded as their physical meaning stop being sensible in case of failure:

- **Loss of TDP collapses all downstream branches**. In case Turbo-Driven Pumps are not available right after the initiating event has taken place, a quick dry-out of the steam generator may occur. This would lead to lose the major heat sink to the secondary, ending up in core uncovery before any following event;

- **Large leak is logically disjoint from no small leak**. TE3 represents the 4 amplification of the existing small seal leak when saturation is reached at the RCP. If TE1 = success (no seal leak ever develops), there is no leak to amplify. The two events are not independent. The Large Leak top event is therefore not branched in the upper half of the tree, and the path goes directly from AFW & TDP success to Extra Batteries. This eliminates 4 phantom sequences that have no physical meaning.

4.2 Expected Results

This section presents the results we expect based on the expected physical behavior of the plant. Each sequence is derived taking into account the success criteria of each TE and the most likely outcome in case of failure or success. They can be seen in Figure 18. The lower half of the picture represents the possible outcomes in case of small leak (TE1) occurs, while the upper part represents the other possible path. As we said before, is TE1 = success, which implies no small leak occurring, a large leak is not going to amplify an already existing break (cause there is none) and therefore the two events are not indipendent. This "disjoint" is indicated at the corresponding place within the graph.
In case extra batteries fail, together with no recover of the AC power, we expect the system to have core damage, as it would not be possible to cool the core. Moreover, if extra batteries are available but no AC power is recovered, core damage is expected to take place as the batteries would get depleted while operating. The unavailability of the Auxiliary Feedwater System and Turbo-Driven pumps would lead to core damage as the main heat sink present within the plant - the steam generators - would be lost.

At the lower part, the sequences are extrapolated following the same scheme applied on the upper part. Core damage will occur if no AFW & TDP are available. Depending on the availability of the extra batteries and the capability of recovering AC power, the predicted outcomes are drawn.
The RELAP analysis will be used either to confirm this configuration or assess different types of events and outcomes.

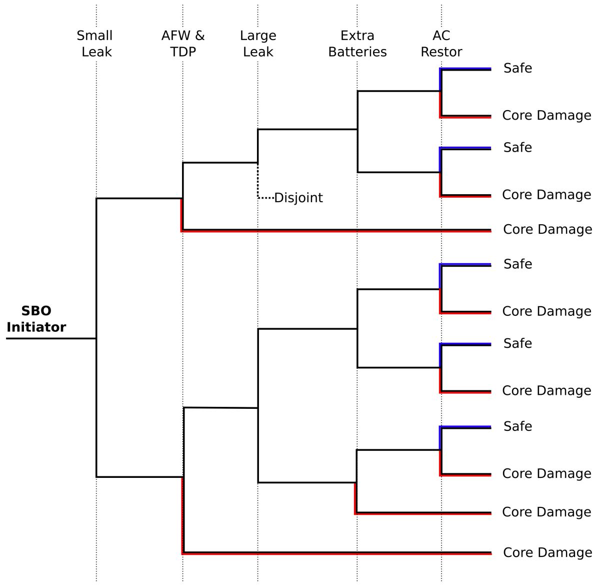

Figure 17: Expected Event Tree.

4.3 Results

Table V: Results of the simulated sequences: Y/N indicates success/failure. Core Damage is obtained when the peak cladding temperatures is higher than 1477K (1204 °C).

Seq.	**TE1** SL	**TE2** AFW	**TE3** LL	**TE4** Batt	**TE5** AC	**Outcome**	**PCT [K]**	t_{CD} **[h]**
1	Y	Y	Y	Y	Y	SAFE	621	—
2	Y	Y	Y	Y	N	CD	1702	22.52
3	Y	Y	Y	N	Y	SAFE	621	—
4	Y	Y	Y	N	N	CD	1701	18.35
5	Y	N	Y	Y	Y	SAFE†	963	—
6	N	Y	Y	Y	Y	SAFE	621	—
7	N	Y	Y	Y	N	CD	1701	21.19
8	N	Y	Y	N	Y	SAFE	621	—
9	N	Y	Y	N	N	CD	1701	17.39
10	N	Y	N	Y	Y	SAFE	621	—
11	N	Y	N	Y	N	CD	1701	21.32
12	N	Y	N	N	Y	SAFE	621	—
13	N	Y	N	N	N	CD	1701	17.42
14	N	N	N	Y	Y	CD	1478	12.21
15	N	N	N	Y	N	CD	1701	12.29

Table V presents the RELAP simulation's results. Several considerations can be made regarding the difference between expected and actual results:

- AC and AFW recovery is the dominant factor that determines whether core damage will occur or not. In all the scenarios when AC and AFW are available, the PCT does not overcome 621 K. The combination of AFW and AC recovery is sufficient to prevent core damage. Once AC power is restored, HPIS and LPIS can keep on cooling the core.

- Sequence 5 (ET5) shows that, even though the AFW is not available, in a condition without leakages through the pump's seal the batteries are able to avoid core damage until the AC power is recovered. This can be explained taking a look at the PCT: without AFW, the temperature rises until the PORV valve starts operating, releasing steam from the pressurizer in order to depressurize the primary. This is different from what we have considered in the preliminary event tree, where the unavailability of AFW determines core damage indipendently from the other systems availability.

- The amplification of the leak, that takes place when the water reaches saturation temperature, has a small effect on the outcome of that sequence. For instance, if we compare ET07 to ET11 or ET09 with ET13, the difference in time is relatively small. This behavior suggest that the main mechanism that lead to core damage is the heat balance through the steam generators, while the leak from the RCP plays a marginal role in the behavior of the plant.

- ET14 can be considered as a "bordeline" case, since its PCT reaches 1478K, slightly above the safety limit. This demonstrates that in presence of leaks from the seals and no AFW, the batteries are not sufficient to prevent core damage. The time needed to reach PCT is 21 minutes higher than the recovery of the AC, which occurs at 12h after the SBO event.

In Figure 18 all the sequences are plotted, as a function of the peak cladding temperature and the time to reach core damage.

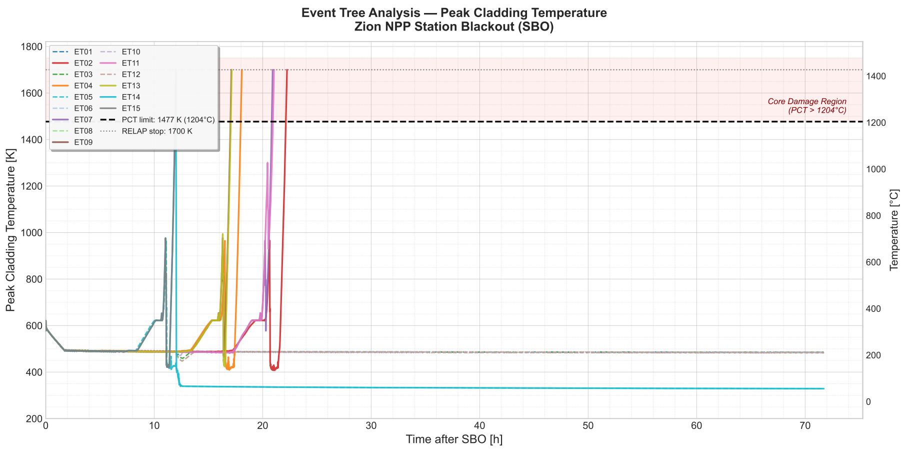

Figure 18: Obtained Event Tree.

5 Probabilistic Safety Analysis

The creation and verification of the event tree for the SBO case allows for the calculation of the contributions of the SBO to the Core Damage Frequency and to identify the Conditional Core Damage Frequency (CCFD).

5.1 Mathematical Representation

The first step to analysis the event tree qualitativly is identify the initating event frequency and the conditional probabilities of each of the elements of the event tree. The key variables under consideration are as follows:

I	Initiating Event (SBO)
LT	Long-term SBO
SB	Short-term SBO
A	AFW & TDP
S	Small Leak
L	Large Leak
R	Restoration of AC

Initiating Event

The initiating event frequency of the SBO is derived from the NUREG CR-7110 Vol 2 [1]. From this report the frequency of Long-Term SBO frequency is identified as 2x10^{-5} pry (per reactor year), and the Short-Term SBO frequency is defined as 2x10^{-6} pry. One interesting finding from this is the higher frequency of long-term SBOs than short term ones. From the frequencies above the initiaing event frequency is derived below:

$$\begin{aligned} F(I) &= F(LT \cup ST) \\ &= 2 \cdot 10^{-5} + 2 \cdot 10^{-6} \\ &= 2.2 \cdot 10^{-5} \text{ pry} \end{aligned} \tag{1}$$

AFW & TDP

The AFW and TDP failure frequency is assumed to be largely driven by the failure of the TDP (try to find a source to back up this claim). As such, the probability of failure of this system is taken from the frequency of failure of the TDP system. This system has two failure modes: failure on demand and failure on run. To use a conservative estimate, the sum of the probability of the failure on demand and failure during a mission time of 5 hours is used. The failure on demand probability is:

$$P(A_d) = 4.6 \cdot 10^{-3}$$

The failure on run probality is calculated as follows:

$$\begin{aligned} P(A_r) &\approx \lambda_f \cdot t_m \\ &\approx 8.45 \cdot 10^{-5}\text{hr}^{-1} \cdot 5\text{hr} \\ &\approx 4.225 \cdot 10^{-5} \end{aligned} \tag{2}$$

Giving a total probability of:

$$\begin{aligned} P(A) &= P(A_d \cup A_r) \\ P(A) &= 4.54\text{x}10^{-5} \end{aligned} \tag{3}$$

Small Leak

The small leak event is assumed to happen in the case that the Westinghouse Shut Down Seal. A primilinary safety analysis of this component from Westinghouse gives it a conservative failure frequency of 1 in 100 cases giving us the follow probability:

$$P(S) = 0.01$$

Large Leak

Large leaks in the RCP seals occur in the situation in which saturation temperatures are reached. The conditional probability of this event during an SBO is listed in the NUREG CR-7110 Vol. 2 report [1], it gives a the following conditional proability.

$$P(L) = 0.2$$

The conditional probability is used in this case as the probability of a large leak dramatically increases during an SBO event.

AC Power Restoration

The AC power restoration after 12 hours is estimating using the conditional probability of a long term SBO given an SBO event. This method allows us to obtain the probability that AC power is restored within a 12 hour window. The calculation is done using Bayes Theroum for conditional probabilities:

$$\begin{aligned} P(R) &= P(LT|I) \\ &= \frac{P(I|LT) * P(LT)}{P(I)} \\ &= \frac{1(2\text{x}10^{-5})}{2.2\text{x}10^{-5}} \\ &= 0.909 \end{aligned} \tag{4}$$

subsectionCCDP
The conditional core damage frequency is the probability of core damage given an intiaing event happens. This allows for the analysis of the consequence that an event can have even if the event is exceedingly rare. To begin the analyssi, a boolean representation of the event tree is made using the sum of the Minimal Cut Sets followed by the simplification of the equation. The minimum cut sets are the paths that lead to core damage and are taken from Figure 18. The equation begins as follows:

$$\begin{aligned} P(CD|I) = {} & P(S)\,P(A)\,P(E)\,\overline{P(R)} \\ & + P(S)\,P(A)\,\overline{P(E)}\,\overline{P(R)} \\ & + P(S)\,\overline{P(A)} \\ & + \overline{P(S)}\,P(A)\,P(L)\,P(E)\,\overline{P(R)} \\ & + \overline{P(S)}\,P(A)\,P(L)\,\overline{P(E)}\,\overline{P(R)} \\ & + \overline{P(S)}\,P(A)\,\overline{P(L)}\,P(E)\,\overline{P(R)} \\ & + \overline{P(S)}\,P(A)\,\overline{P(L)}\,\overline{P(E)} \\ & + \overline{P(S)}\,\overline{P(A)} \end{aligned} \tag{5}$$

This equation is then simplified using boolean algebra identifies to get the final CCDF equation below:

$$\begin{aligned} P(CD|I) &= \overline{P(A)} + P(S)P(A)\overline{P(R)} + \left(\overline{P(E)} + P(E)\overline{P(R)}\right)\left(\overline{P(S)}P(A)\overline{P(L)}\right) \\ &= 4.64\text{x}10^{-3} + 0.089 + ()(0.002) \\ &= \ldots \end{aligned} \tag{6}$$

5.2 Core Damage Frequency

$$F(CD) = F(I)\ P(CD|I) \tag{7}$$

6 Preliminary Conclusions

From the first analyses, the system code RELAP5 can be used as the model basis for the next stages of the project. The steady-state calculation is first checked against the target values, and the SBO base case is then used to observe the main thermal-hydraulic response of the plant under blackout conditions.

The SBO base case transient showed that the plant is able to maintain core cooling for at least 12 hours without core damage. The decay heat is effectively removed through natural circulation in the primary system and by the Auxiliary Feedwater system on the secondary side. The peak cladding temperature remains well below the safety limit of 1477 K throughout the entire transient, and the core is never uncovered.

Having analyzed two EOPs, we can say that the SG cooldown procedure (EOP 1) proved usefil but not strictly necessary to prevent core damage within the 12-hour simulation window. The cooldown EOP reduces primary pressure and lowers the RCP seal leak rate. It is therefore recommended as a prudent operator action. Instead, the primary depressurization procedure (EOP 2) was never triggered in either the base case or the no-cooldown case, as the Core Exit Temperature did not reach the 923 K threshold at any point during the transient. Overall, the results demonstrate that the existing safety systems and operator procedures are sufficient to manage the SBO scenario for 12 hours. However, the analysis also highlights that the absence of the SG cooldown EOP leads to a more challenging scenario and could lead to core damage in a longer event.

At this stage, the report is intended as a structured first draft. The final discussion will be completed after adding the detailed numerical results, the transient plots, and the comparison with the additional scenarios required in the following stages of the project.

CRediT Author Statement

- **Filippo Dalmonego:** Coding, Writing, Review, Supervision, Visualization.
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References

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